



Miss Mehwish: 03171491481

PHY101

Final Term (Live Quiz)

PAID VU LMS HANDLING by Mam Mehwish

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1. Which of the following statements about floating objects is correct?

- The object's density is greater than the density of the fluid on which it floats.
- The displaced volume of fluid is greater than the volume of the object.
- The object's density is equal to the density of the fluid on which it floats.
- The buoyant force equals the object's weight.

2. Suitable units for the gravitational constant G are:

- $\text{kg}\cdot\text{m}/\text{s}^2$
- $\text{N}\cdot\text{m}^2/\text{kg}^2$
- $\text{m}^3/(\text{kg}\cdot\text{s}^2)$
- m/s^2

3. Imagine a hypothetical scenario where shock waves could be produced.

- A gentle breeze blowing across a pond.
- A person clapping their hands.
- A supersonic aircraft breaking the sound barrier.
- A flashlight producing light waves.

4. A thermometer indicates 98.6°C . It may be:

- Outdoors on a cold day



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- In a comfortable room
- In a cup of hot tea
- In a normal person's mouth

5. The direction of angular momentum of a body moving in a circle is:

- Perpendicular to the plane of circle
- Radially inward
- Radially outward
- Along the tangent

6. Hooke's Law states that the force in a spring is proportional to:

- The equilibrium length
- The mass attached
- The square of displacement
- The displacement from equilibrium

7. The equation for time dilation is given by:

- $t' = \gamma t$
- $t = \gamma t'$
- $t = t'$
- $t = \gamma t'$

8. In an adiabatic process:

- The energy absorbed as heat equals the work done by the environment on the system
- The energy absorbed as heat equals the work done by the system on its environment
- The work done by the environment on the system equals the change in internal energy
- The absorbed as heat equals the change in internal energy

9. A net torque applied to a rigid object always tends to produce:

- linear acceleration



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- rotational inertia
 - angular acceleration
 - rotational equilibrium
1. The angular velocity of an object in circular motion is doubled while maintaining a constant radius, how does the centripetal acceleration change?
 - It doubles ☒
 - It quadruples
 - It becomes half
 - It remains the same
 2. The momentum of an object at a given instant is independent of its:
 - velocity
 - mass
 - speed
 - acceleration ☒
 3. If electric and gravitational force on an electron in a uniform electric field balance each other, then the intensity of electric field will be:
 - q/mg
 - m/ag
 - mg/q ☒
 - qg/m
 4. A rod with moment of inertia of 5 kgm^2 and oscillates with a period of 3.14 seconds, its torsion constant "k" is:
 - 15 Nm
 - 0.62 Nm
 - 20 Nm



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- 10 Nm ☒
- 5. A man, with his arms at his sides, is spinning on a light frictionless turntable. When he extends his arms:
 - his angular velocity increases ☒
 - his rotational inertia decreases
 - his angular velocity remains the same
 - his angular momentum remains the same
- 6. In a frictionless environment, a mass oscillates horizontally in a mass-spring system. What are the energies involved in it?
 - Kinetic and gravitational potential energies
 - Gravitational potential and elastic potential energies
 - Impossible to determine
 - Kinetic and elastic potential energies ☒
- 7. If the inclined plane is not frictionless, how does the linear velocity at the bottom of the incline for the solid disk compare to that of the hoop?
 - The linear velocity depends on the mass of the objects
 - Both have the same linear velocity
 - The linear velocity of the hoop is greater
 - The linear velocity of the solid disk is greater ☒
- 8. Alan ----- is the basic reason to change in the motion of an object according to Newton's second law of motion.
 - decrease in inertia
 - acceleration
 - net force ☒
 - change in velocity
- 9. If the pulley in the system has mass and friction, which factor would most likely affect the tension in the rope?



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- The color of the pulley.
- The temperature of the room.
- The material of the pulley.
- The mass of the pulley. ☒

10. What is the angular speed of the second hand in an analog clock?

- 360 degrees per minute ☒
- 6 degrees per minute
- 360 degrees per hour
- 30 degrees per hour

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Question 1: How does the length of a simple pendulum (L) affect its period?

The period is inversely proportional to the square of the length.

It affects the period directly ✓

It doesn't affect the period

It affects the period indirectly

Question 2: Instantaneous velocity is calculated by:

Average distance covered over time

Displacement over average velocity

Total distance divided by total time

Slope of the displacement-time graph ✓

Question 3: Which of the following will remain unchanged when a sound wave travels in air or in water?

Amplitude



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Wavelength

Speed

Frequency ✓

Question 4: What is the dimension of angular velocity?

[T-1] ✓

[T]

[LT-2]

[LT-1]

Question 5: What is a physical quantity that has both magnitude and direction called?

Constant

Variable

Vector ✓

Scalar

Question 6: What force does a liquid exert on a line that is two meters in length if its surface tension is 20 N/m?

40 N ✓

20 N

10 N

30 N

Question 7: If light of low wavelength is used in Young's double slit experiment, then width of the fringe will:

Decrease ✓



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No change

Double

Increase

Question 8: What is the correct expression for relativistic momentum $\vec{p}$?

$\vec{p} = \gamma m \vec{u}$ ✓

$\vec{p} = m \vec{u}$

$\vec{p} = \gamma m \vec{u} \text{ ---- } \checkmark$

$\vec{p} = m \vec{u} \gamma$

Question 9: Which system of units uses the foot, pound, and second as its fundamental units?

MKS

SI

CGS

FPS ✓

Question 10: Due to relative motion of observer and the frames of reference of events, time always:

Stays itself

Stretches itself

Continuous itself

Contracts itself ✓

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Question 1: Let M denote the mass of Earth and let R denote its radius. The ratio g/G at Earth's surface is:



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MR^2

M/R^2 ✓

M/R

R^2/M

Question 2: If you double the height of an object above the Earth's surface, how does the gravitational potential energy change?

It halves

It doubles ✓

It becomes zero

It remains constant

Question 3: The two metallic strips that constitute some thermostats must differ in:

Coefficient of linear expansion ✓

Thickness

Length

Rate at which they conduct heat

Mass

Question 4: If the mass 'm' and the spring constant 'k' are increased by a factor of 2, how does the angular frequency 'ω' of a simple pendulum change?

It increases

It becomes zero

It remains same ✓

It decreases

Question 5: Which system of units uses the foot, pound, and second as its fundamental units?



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CGS

SI

FPS ✓

MKS

Question 6: A whistle producing sound waves of frequencies 9500 Hz and above is approaching a stationary person with speed v m/s. The velocity of sound in air is 300 m/s. If the person can hear frequencies up to 10000 Hz, the maximum value of v up to which he can hear the whistle is:

1.2 ✓

1.45

1.5

1.4

Question 7: What is the relationship between external torque (T) and angular acceleration (α)? Where, F = Force, I = moment of inertia, W = work done, a = acceleration, m = mass.

$T = F\alpha$

$T = IF$

$T = m\alpha$

$\alpha = T/I$ ✓

Question 8: The transformer voltage induced in the secondary coil of a transformer is mainly due to:

A varying electric field

The vibrations of the primary coil

The iron core of the transformer

A varying magnetic field ✓



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Question 9: A car of mass 1000 kg accelerates from rest to 20 m/s in 10 seconds. What is the average force applied?

1000 N

200 N

3000 N

2000 N ✓

Question 10: The angular displacement of one revolution is equal to:

$\pi/2$ radian

1 radian

π radian

2π radian ✓

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Momentum is a vector quantity defined by:

Mass times velocity squared

Force divided by velocity

Mass times velocity ✓

Mass times force

2. What is the value of gravitational potential energy of an object when it is placed at infinite height?

It decreases

It remains constant

It depends on the object's shape



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It increases ✓

3. When an electromagnetic wave meets a reflecting surface, the direction taken by the reflected wave is determined by:

The angle of incidence ✓

1.

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The index of the medium

The intensity of the wave

The material of the reflecting surface

4. In a series circuit with resistances R_1 and R_2 , if R_1 is much greater than R_2 , what is the approximate total resistance (R)?

$$R = R_1 + R_2$$

$$R = R_1$$

$$R = R_2$$

$$R = R_1 - R_2 \quad \checkmark$$

5. In the electricity standards are:

220 Volts and 50 Hertz ✓

100 Volts and 60 Hertz

120 Volts and 60 Hertz

120 Volts and 50 Hertz

6. What happens if a point source is placed at the focus of a concave lens?



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Parallel beam emerges ✓

Rays scatter

Rays converge to a point

Rays reflect back

7. The center of mass of a uniform disk of radius R is located:

on the rim

at a distance $R/2$ from the center

at a distance $R/3$ from the center

at the center ✓

8. Which of the following statements are TRUE of sound? (Identify all that apply.)

A sound wave is a transverse wave

Sound can travel through a vacuum

To hear the sound of tuning fork, the tines of the fork must move or run back and forth one's ear

A sound wave is a mechanical wave ✓

9. In magnetic field, which ions experience a greater deflection (ions are heavier or lighter)? (Provided all other parameters are kept same)

Lighter ions ✓

Heavier ions

Independent of ion-mass

Both experience same deflection

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Note: Live quiz MCQs also repeated in Exam (6, 7)



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